Introduction to the Kernel and Device Drivers

Introduction

- Device drivers
  - Software interface to hardware device
  - Use standardized calls
    - Independent of the specific driver
  - Main role
    - Map standard calls to device-specific operations
  - Can be developed separately from the rest of the kernel
    - Plugged in at runtime when needed

Role of the Device Driver

- Implements the *mechanisms* to access the hardware
  - E.g., Hard disk
    - Read/write blocks of data
    - Access as a continuous array
- Does not force particular *policies* on the user
  - Examples
    - Who many access the drive
    - Whether the drive is accessed via a file system
    - Whether users may mount file systems on the drive

Policy-Free Implementation

- Simplifies the design
- Separation of concerns
  - Capabilities provided
  - Use of Capabilities
- Reuse
  - Different policies do not require changes to the kernel

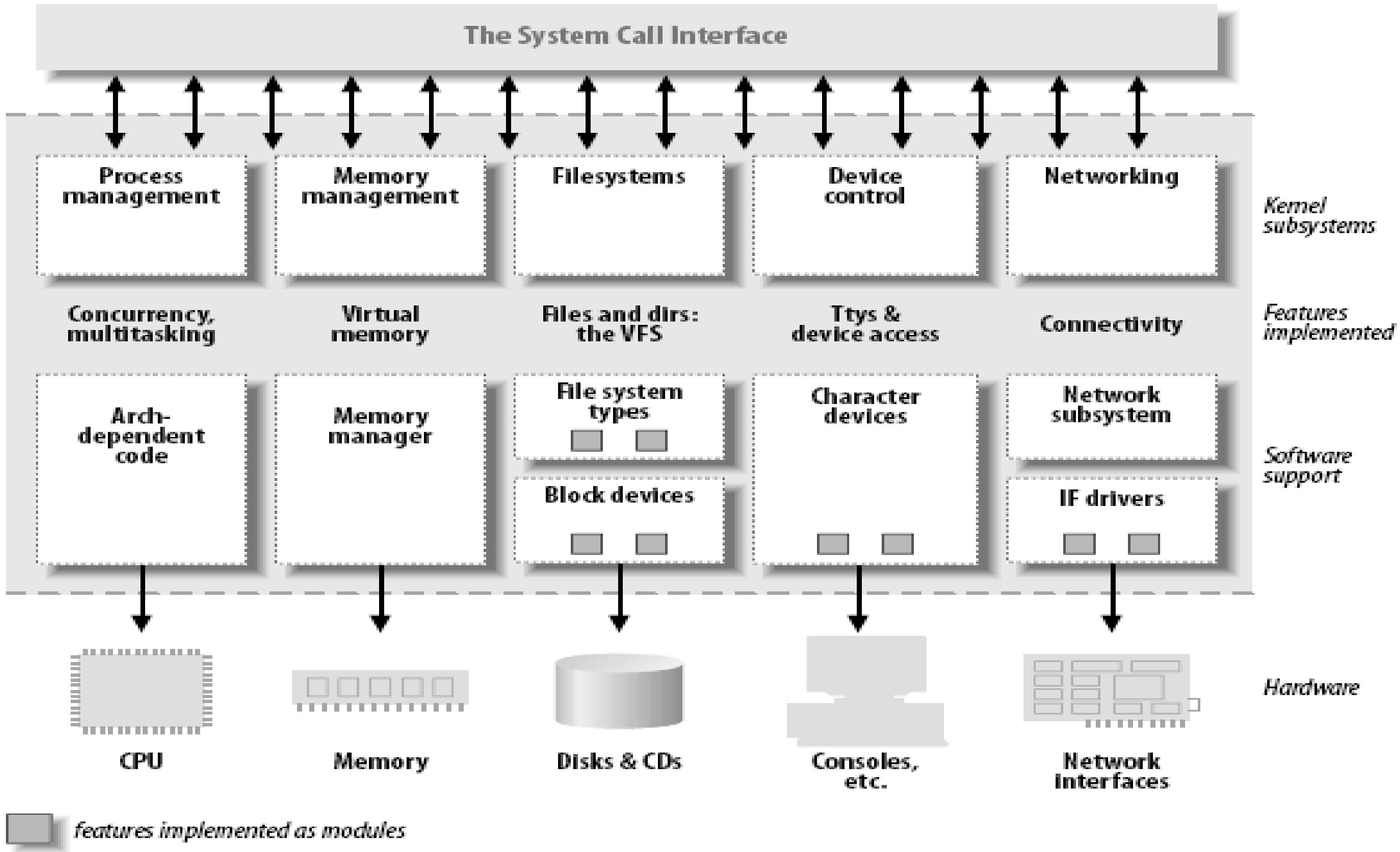
Splitting the Kernel

- Kernel handles resource requests
- Process management
  - Creates, destroys processes
  - Supports communication among processes
    - Signals, pipes, etc.
  - Schedules how processes share the CPU/cores
- Memory management
  - Virtual addressing

Splitting the Kernel

- File systems
  - Everything in UNIX can be treated as a file
  - Linux supports multiple file systems
- Device control
  - System operation maps to a physical device
- Networking
  - Handles packets
  - Routing and network address resolution

Splitting the Kernel



(Dynamically) Loadable Modules

- The ability to add and remove kernel features at runtime
- Each unit of extension is called a *module*
- Each module is made up of object code that can be dynamically linked to the running kernel by the **`insmod and modprobe`** program to add a kernel module
- Use **`rmmod`** program to remove a kernel module

Classes of Devices and Modules

- Character devices
- Block devices
- Network devices
- Others

Character Devices

- Abstraction: a stream of bytes
  - Examples
    - Text console (**`/dev/console`**)
    - Serial ports (**`/dev/ttyS0`**)
- Usually supports **`open`**, **`close`**, **`read`**, **`write`**
- Accessed sequentially (in most cases)
  - Might not support file seeks
- Accessed through a filesystem node

Block Devices

- A block devices are accessed by filesystem nodes in the (/dev) directory
- Abstraction: continuous array of storage blocks (e.g. sectors)
  - Applications can access a block device in bytes
- Accessed through a file system node
- In most Unix systems, a block device can only handle I/O operations that transfer one or more whole blocks, which are usually 512 bytes in length.
- A block device can host a file system

Network Devices

- Abstraction: data packets
- Send and receive packets
  - Do not know about individual connections
- Have unique names (e.g., **`eth0`**)
  - Not in the file system
  - Support protocols and streams related to packet transmission (i.e., no **`read`** and **`write`**)

Other Classes of Devices

- Examples that do not fit to previous categories:
  - USB
  - SCSI
  - FireWire
  - MTD

File System Modules

- Software drivers, not device drivers
- Serve as a layer between user API and block devices
- Intended to be device-independent

Security Issues

- Deliberate vs. incidental damage
- Kernel modules present possibilities for both
- System does only rudimentary checks at module load time
- Relies on limiting privilege to load modules
  - And trusts the driver writers
- Driver writer must be on guard for security problems

Security Issues

- Do not define security policies
  - Provide mechanisms to enforce policies
- Be aware of operations that affect global resources
  - Setting up an interrupt line
    - Could cause another device to malfunction
  - Setting up a default block size
    - Could affect other users

Security Issues

- Buffer overrun
  - Overwriting unrelated data
- Treat input/parameters with utmost suspicion
- Uninitialized memory
  - Kernel memory should be zeroed before being made available to a user
  - Otherwise, information leakage could result
    - Passwords

Security Issues

- Avoid running kernels/device drivers compiled by an untrusted friend
  - Modified kernel could allow anyone to load a module

Version Numbering

- Every software package used in Linux has a release number
  - Version dependency
    - You need a particular version of one package to run a particular version of another package
  - Prepackaged distribution contains compatible versions of various packages

Version Numbering

- Different throughout the years
  - 1.0 … 2.4 … 2.6… 3.0
- After version 1.0 but before 3.0
  - <major>.<minor>.<release>.<bugfix>
  - Time based releases (after two to three months)
- 3.x
  - Moved to 3.0 to better reflect newer development process
  - <version>.<release>.<bugfix>
  - https://lkml.org/lkml/2011/5/29/204

License Terms

- GNU General Public License (GPL2)
  - GPL allows anybody to redistribute and sell a product covered by GPL
    - As long as the recipient has access to the source
    - And is able to exercise the same rights
  - Any software product derived from a product covered by the GPL must be released under GPL

License Terms

- If you want your code to go into the mainline kernel
  - Must use a GPL-compatible license

Joining the Kernel Development Community

- The central gathering point
  - Linux-kernel mailing list
    - http://www.tux.org/lkml
    - https://lkml.org/
- Chapter 20 of LKD Book further discusses the community and accepted coding style